

Facial Gesture Recognition Using Bio-impedance Sensing

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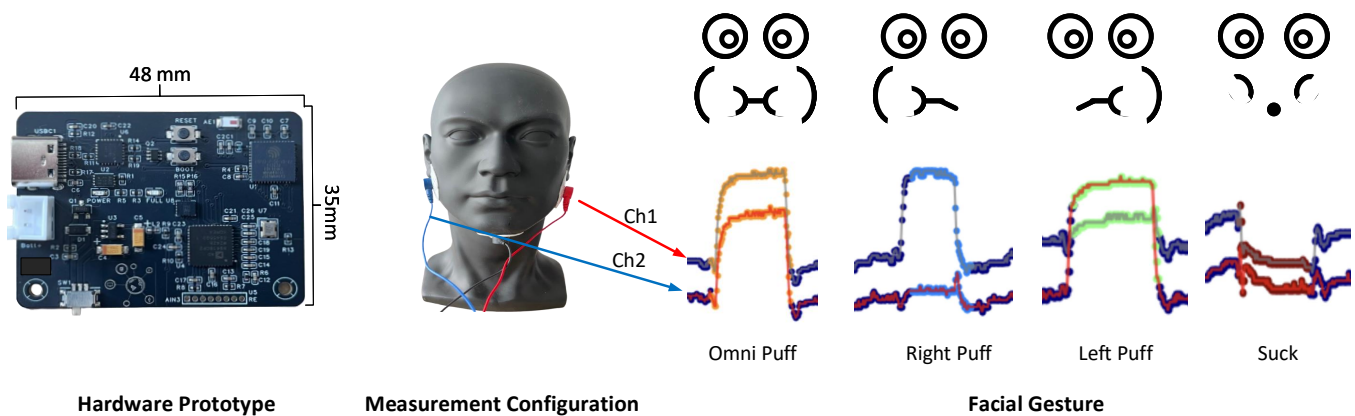


Figure 1: Facial gesture recognition using bio-impedance sensing. The wearable hardware prototype is built based on the AD5941 bio-impedance sensing module and the microcontroller ESP32PICO. Four puff cheek gestures are detected by the proposed method. The raw signals (only magnitude is plotted here) from the two bio-impedance channels are presented under each facial gesture. It can be observed that the cheek movement can cause an obvious bio-impedance variation.

Abstract

Facial gesture recognition plays a crucial role in human-computer interaction (HCI), enabling intuitive and non-intrusive input methods. In this work, we present a facial gesture recognition solution based on bio-impedance sensing. Our proposed method effectively detects four types of cheek puff activities and demonstrates its feasibility through an experiment involving four participants performing these gestures. With a Random Forest classifier, our approach achieves an average accuracy of 93 %, highlighting its potential as an input modality in HCI applications.

Keywords

bio-impedance sensing, facial gesture recognition



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1 Introduction

Human-computer interaction (HCI) has seen tremendous progress with the development of innovative input methods designed to enhance accessibility and user experience. While traditional interaction techniques, such as touchscreens, keyboards, and voice commands, are widely used, they are not always practical in certain scenarios. For example, hands-free control may be required when users' hands are occupied, or voice commands may fail in noisy environments [2]. In such cases, facial gestures provide a promising alternative for interaction. They are natural, intuitive, and require minimal effort to perform [3, 4, 6, 14]. However, despite their potential, achieving efficient and accurate facial gesture recognition remains a significant challenge.

Most existing research on facial gesture recognition relies on vision-based approaches that use deep learning and computer vision techniques to track facial expressions and muscle movements [4, 6, 14]. While these methods have demonstrated effectiveness, they come with limitations. Vision-based systems often require high computational resources, struggle in low-light conditions or with occluded faces, and may raise privacy concerns. Alternative sensing modalities leveraging body's electrical field [1], such as electromyography (EMG) and proximity sensing [3, 4, 7], have also been explored for detecting facial activity. EMG-based methods, for instance, measure electrical signals generated by facial muscles using surface electrodes. Although EMG provides high temporal resolution and directly captures muscle activation, it is prone to motion artifacts, depends heavily on precise electrode placement, and often requires skin preparation to ensure signal reliability.

Bio-impedance sensing has recently emerged as a promising solution for human activity recognition (HAR). As a kind of electrophysiological signal, it can monitor the conductive information of the object located between two electrodes (two-electrode configuration). This technology has been researched in many fields, such as respiratory monitoring [13], sports activity recognition [9], and muscle activity detection [5]. More recently, researchers have begun exploring its potential for wearable human activity recognition tasks, including hand gesture [8, 12] and dietary classification [10]. When compared to EMG-based approaches, bio-impedance sensing provides additional benefits. Unlike EMG signals, which are highly sensitive to motion artifacts and require precise electrode placement, bio-impedance measures changes in tissue conductivity between the electrodes, which makes it less susceptible to variability caused by electrode positioning and further provides a more stable signal with fewer motion-related disturbances. Additionally, bio-impedance sensors can detect deeper tissue activity beyond surface-level muscle movements, offering richer information for gesture recognition tasks. Facial gestures often lead to the conductance variation of the face, for example, in puff cheeks, the bio-impedance will be increased as the airflow into the mouth. These advantages make bio-impedance sensing a potential and promising solution for facial gesture recognition in real-world HCI applications. However, bio-impedance sensing-based facial gesture recognition is rarely studied compared to other sensing modalities.

In this work, we demonstrate a novel facial gesture recognition system based on bio-impedance sensing that is capable of detecting three distinct cheek puff activities. To evaluate our approach, we conducted experiments with four participants performing these gestures and achieved an average accuracy of 93% using a random forest classifier. Our findings demonstrate the potential of bio-impedance sensing as an effective input method for HCI applications, offering a lightweight, privacy-preserving, and environment-independent alternative to existing solutions.

2 Related Work

In recent years, several sensor-based facial gesture recognition solutions have been proposed [3, 4, 7, 11]. Li et al. [7] introduced a cost-effective cheek-sensing method to infer continuous jaw motion in mobile virtual reality (VR) applications. Their novel approach employs five VCNL4020 fully integrated proximity sensors with

infrared emitters to track jaw and lip movements, enhancing VR interaction. Gjoreski et al. [4] developed a wearable surface electromyography (sEMG) device to monitor facial muscle activity associated with emotional states. By capturing muscle activation patterns, this device offers an accessible and user-friendly method for tracking affective states in daily life, with potential applications in psychology, human-computer interaction (HCI), and healthcare. Matthies et al. [11] proposed an innovative in-ear electric field sensing technique, dubbed EarFS, to detect facial expressions. Using electrodes placed in the ear canal, EarFS captures electrical changes induced by facial muscle movements, providing an unobtrusive solution for gesture-based input in wearable devices and enabling hands-free interaction in mobile computing environments.

While bio-impedance sensing has shown promising results in human activity recognition (HAR) tasks [9, 10, 12, 13], its application to facial gesture recognition remains underexplored. Therefore, in this work, we investigate the performance of bio-impedance sensing for facial gesture recognition, with a specific focus on cheek movements. Additionally, a related study, GazePuffer [6], addressed a limitation of traditional gaze input by incorporating a confirmation mechanism triggered by cheek puffs. However, that approach still relied on camera-based perception to detect cheek movements, highlighting a gap that our bio-impedance-based solution aims to fill.

3 Methodology

3.1 Hardware Implementation

In this study, we develop a wearable device to measure facial bio-impedance variations induced by cheek puff activities, as illustrated in Figure 1. The core component of the device is the AD5941 bio-impedance sensing module, interfaced with the face via wet electrodes. This component generates a 100 kHz alternating current (AC) signal with a peak-to-peak amplitude of 100 mV, which is applied to the target area. The resulting response current signal is measured through the face, processed by the on-chip hardware Fast Fourier Transform (FFT) accelerator, and the impedance data is retrieved via the SPI interface by the ESP32-PICO-V3 microcontroller, which also supports WiFi and Bluetooth connectivity. The data sampling rate is set to 20 Hz. To ensure safety, the current injected into the body adheres to IEC 60601 standards, which impose limits on both direct current (DC) and alternating current (AC) entering the human body. Specifically, the DC component must be less than 10 μ A, while the maximum allowable AC current is 500 μ A at 50 kHz and 600 μ A at 60 kHz. To comply with these requirements, a 470 nF capacitor and a 1000-ohm resistor were sequentially integrated into the measurement circuit to block DC signals and limit the current flow respectively.

3.2 Experiment

To evaluate the performance of the bio-impedance sensing solution for facial activity recognition, we recruited four volunteers (two females and two males) to participate in the experiment. These participants were asked to wear the proposed hardware and perform four distinct cheek gestures, such as left puff, right puff, omni puff, and suck, as illustrated in Figure 1. To capture two-channel bio-impedance signals, three electrodes were strategically placed on

Table 1: Result Summary

Metric/Subject	1	2	3	4	Average
Accuracy	0.96	0.90	0.97	0.91	0.93
Macro Recall	0.95	0.83	0.96	0.92	0.91
Macro F1-score	0.95	0.87	0.97	0.92	0.92
Macro Precision	0.96	0.92	0.98	0.91	0.94

the face: one electrode was positioned beneath the mouth as the stimuli signal electrode, while the other two were affixed to each cheek as the measurement electrodes, as depicted in Figure 1.

3.3 Data Processing

As shown in Figure 1, facial gestures induce noticeable variations in bio-impedance signals. For example, when the user performs the omni cheek puff, both bio-impedance channels exhibit a significant increase due to the expansion of air volume in the oral cavity. In contrast, sucking gestures result in a clear decrease in bio-impedance as air is expelled. Additionally, a single-side puff primarily affects the signal from one measurement channel. These distinct signal variations enable the classification of four facial gestures using a lightweight algorithm.

The data processing pipeline involves several key steps to ensure effective activity classification. First, to reduce noise, a peak-removal method is applied by replacing extreme values with the mean. A sliding window approach (window size = 20) is then employed to segment the data into overlapping feature vectors, where each window concatenates 20 consecutive readings from both impedance channels. The label assigned to each window corresponds to the last time step in the segment. The processed data is subsequently split into training (80%) and testing (20%) sets. A Random Forest classifier, consisting of 100 decision trees, is trained on these features to learn activity patterns. Finally, model performance is evaluated using a classification report (accuracy, precision, recall, and f1 score), providing insights into the predictive capability of the trained model.

4 Results

Table 1 shows the facial gesture classification result summary. The classification results indicate strong overall performance. The model achieved an average accuracy of 93%, demonstrating its reliability across subjects. The macro recall (91%) suggests that the model performs well in identifying all classes, though subject 2 had a relatively lower recall (83%), indicating some misclassification. The macro F1-score (92%) reflects a good balance between precision and recall, confirming the model’s robustness. Overall, the classifier shows strong and consistent performance across different subjects, with slight variability in recall.

5 Conclusion

In this study, we showcased the promising potential of bio-impedance sensing for facial gesture recognition. Using a two-channel bio-impedance signal collected from the face, we successfully identified four distinct cheek movements and null class with a random forest classifier, achieving an average accuracy of 93%. This high accuracy

underscores the potential of bio-impedance sensing as a robust method for facial gesture recognition, with applications in various scenarios in the field of HCI, where it can serve as an effective and alternative hands-free input technique.

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